

Venkata Sai Sreeharsha Kotta

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SUMMARY

Machine Learning Engineer with 5+ years of experience productionizing ML, deep learning, and LLM systems at enterprise scale. Strong in data engineering with **Databricks** and **PySpark**, building and optimizing training and serving pipelines, feature engineering, and model tuning for reliability and performance in production. Experienced across AWS and Azure cloud data services, streaming, CI/CD, and observability, with a track record of translating business requirements into scalable, maintainable ML solutions with quantifiable value.

EDUCATION

- **Georgia State University** Atlanta, USA
Master of Science in Analytics; Data Science and Analytics; GPA: 4.0/4.0 Aug. 2022 - May 2024
- **Shiv Nadar University** Delhi, India
Bachelor of Technology in Computer Science and Engineering; Minor in Management Aug. 2016 - Jun. 2020

EXPERIENCE

- **Vinci Education Corporation** Dunn Loring, VA, USA
Senior Machine Learning Engineer May 2024 – Present
 - Led end-to-end design and **productionization** of a real-time detection platform scaled to **100+ edge devices and 600+ cameras** across multiple sites, using an **edge-first MicroK8s** architecture for on-device inference and **AWS EKS** for the web app, messaging/streaming, and operational workflows.
 - Designed a multi-model video-understanding pipeline, **evaluating and integrating pretrained and vendor models** (**YOLO**, pose estimation, **3D CNN** RGB/motion, transformer temporal models, and larger **vision-language models/LLMs**) to classify abuse vs non-abuse and generate scene descriptions.
 - Built a **semantic video-search** capability with large vision-language models, enabling natural-language retrieval across all cameras and recorded footage for rapid incident review.
 - Improved model accuracy through **feature engineering, threshold tuning, temporal smoothing, and observability**, processing **~9,000 segments per camera** while surfacing only **1-2 actionable alerts** and sharply reducing false positives
 - Improved **computational performance** across the pipeline by optimizing GPU inference, batching, and pipeline stages on **Ubuntu + NVIDIA GPU**, cutting end-to-end latency from **6-7s to <2s at 25 FPS**.
 - Orchestrated, scaled, and hardened **model training and serving pipelines** with **SageMaker Pipelines, AWS Lambda + S3, model registry/versioning**, and **CI gating tests**, promoting only validated models across all customers and devices.
 - Implemented monthly **auto-retraining loops** with **model monitoring and drift detection** from production feedback, plus a **privacy-preserving face-blurring pipeline** for secure, compliant data handling.
 - Curated and **versioned a 7K+** labeled clip dataset with QA and augmentation, ensuring reproducible training and consistent feature/label governance.
 - Built **CI/CD and GitOps** deployment for edge and cloud using **GitLab CI/CD, Helm + ArgoCD**, and **external-secrets synchronization**, enabling consistent configuration, auditability, and rapid rollback.
 - Operated a cloud platform on **AWS EKS (Spring Boot + Angular + SQL)** with **Nginx** and **cert-manager**, and built the cloud frontend and **S3 data-lake pipelines** for media storage and reporting.
 - Implemented edge-to-cloud **messaging and streaming** with **RabbitMQ** and **VerneMQ** for device communication, deployment, and status updates, ensuring stable connectivity and fail-safe behavior.
 - Set up **observability** with **Prometheus, Grafana, and Loki** (camera activity, pod failures, pipeline health) plus reporting dashboards, and automated site onboarding to cut deployment time from **6-7 hours to <30 minutes** per site; **mentored junior engineers**.
- **Institute of Insight, Georgia State University** Atlanta, USA
Graduate Research Assistant Jan 2023 – May 2024
 - Extracted and analyzed unstructured financial text at scale using Pushshift (multi-thread), PRAW, and web scraping across Reddit, news, and earnings-call transcripts; applied **NLP** to identify company names/tickers and surface supply-chain topics
 - Implemented textual analysis with seeded word embeddings (**Word2Vec, CBOW**) in Python to build an organization-level metric assessing supply-chain ethicality from web-scraped data.
 - Applied **Large Language Models** for content classification, labeling millions of tweets to detect mental-trauma signals at scale.
- **Hewlett Packard Inc** Bengaluru, India
Data Scientist Aug 2020 – Aug 2022
 - Automated end-to-end **ETL, model training, and evaluation** via scalable data pipelines in **Azure Databricks** and **Azure Data Factory** using **Spark (PySpark, SparkR)** on distributed clusters, saving **120 labor hours monthly**.

- Developed **time series forecasting** models with **machine learning and deep learning** in Python and R for Part Demand, Repair, Call Volumes, and Warranty Interventions, improving budget accuracy by **15%** and reducing forecasting errors by **20%**.
- Forecasted spare-parts demand with **89% accuracy**, driving **\$26M savings** by minimizing overstock and part wastage; led global field-service coverage optimization, boosting accuracy by **17%** via a Staff Planning Tool.
- Built a **Spare Part Replacement Recommendation Engine** using **NLP and LLMs** (Transformers) to help customer-service agents assist with damage and repair, reducing wait time to **15 seconds**.
- Applied **feature engineering** and predictive analytics for operational risk and resource-allocation analysis, optimizing supply-chain resources and inventory management.
- Published part-demand forecasting research at HP's internal **DSKD Summit** and delivered insights to cross-functional stakeholders through interactive **Power BI** dashboards.
- **Hewlett Packard Inc** Chennai, India
Data Science Intern *Feb 2020 – Aug 2020*
 - Performed exploratory data analysis on monthly intervention snapshots to support long-term modeling with an expected ROI of **\$6M** over 3 years.
 - Supported statistical and ML-based demand forecasting for inventory optimization and contributed to early pipeline automation using Python and R.

PROJECTS

- **AI-Driven Video Summarization and Deepfake Synthesis:** Combined SSD face detection and Wav2Lip lip-sync with NLP tooling (Gensim, NLTK, spaCy, TF-IDF) for summarization, Tortoise TTS for voice conversion, and Google Speech Recognition; delivered an interactive Streamlit interface with attention to ethical use of deepfake technology.
- **Marketing Budget Optimization using Markov Chains:** Used R and the Channel Attribution package to model attribution paths and optimal budget allocation, comparing Markov Chains with heuristic models to improve channel evaluation and campaign performance.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Scala, R, Java, Bash, PySpark, SparkR
- **Data Engineering & Distributed:** Azure Databricks, Azure Data Factory, Spark (PySpark, SparkR), Data Lakes (S3), ETL & Feature Pipelines, Streaming, REST APIs
- **Machine Learning & DL:** Regression, Classification, Ensemble Methods, Clustering, Feature Engineering, Model Optimization & Parameter Tuning, Model Evaluation (Precision/Recall/F1/ROC-AUC), Time Series Forecasting, NLP, GenAI/LLM Fine-tuning, CNNs, Transformers
- **Libraries & Frameworks:** PyTorch, TensorFlow, Scikit-Learn, OpenCV, HuggingFace Transformers, Whisper, LangChain, Pandas, NumPy, Matplotlib, Seaborn
- **MLOps & Deployment:** Docker, Kubernetes (MicroK8s, EKS), Helm, ArgoCD (GitOps), SageMaker Pipelines, Model Registry/Versioning, GitLab CI/CD, Git, AWS ECR/Lambda, External Secrets
- **Cloud Platforms:** AWS (EC2, EKS, S3, Lambda, SageMaker, ECR, CloudWatch, Systems Manager), Azure (Databricks, ADF)
- **Databases & Streaming:** PostgreSQL, MySQL, MongoDB, DynamoDB, RabbitMQ, VerneMQ
- **Monitoring & Visualization:** Prometheus, Grafana, Loki, Power BI, Tableau, Excel
- **Practices:** Agile, SDLC, MLOps, Feature Engineering, Model Governance

PUBLICATION

- Object and Currency Detection with Audio Feedback for Visually Impaired. In IEEE Region 10 Symposium (TENSYP), pp. 1152-1155. IEEE, 2020

CERTIFICATIONS

Deep Learning Specialization – Coursera (2020) | Natural Language Processing Specialization – Coursera (2020) | Machine Learning – Coursera (2019)